Probabilistic Listener: *A Case of Reflexive ziji "self" Ambiguity Resolution in Mandarin*

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INTRO

- What mechanisms do people use to resolve ambiguous pronouns?=> **Bayesian models** [1]
- **Ambiguous reflexive *ziji* in Mandarin:** permits local and non-local referents.

[小红]$_i$ 说 [小明]$_j$ 总 把 [自己]$_{i/j}$ 弄糊涂
[Hong]$_i$ says [Ming]$_j$ always BA **[ziji]**$_{i/j}$ confuses.
[Hong]$_i$ says that [Ming]$_j$ always confuses **[self]**$_{i/j}$.

- Previous studies ask this question using cross-sentential pronouns [2]. **What about within-sentence pronoun dependencies?**

METHODS

- Two Bayesian models:

1. Simple Bayesian Model (SBM) [3]:

$$P(\text{ref} \mid \text{utt}) \propto P(\text{ref}) \times P(\text{utt} \mid \text{ref})$$

2. Rational Speech Act (RSA) [4]:

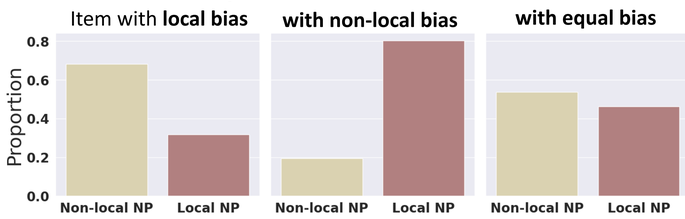
$$P(\text{ref} \mid \text{utt}) \propto P(\text{ref}) \times \text{Rational Speaker}$$

$$\text{Rational Speaker} = \text{Literal listener} - \text{Cost}(\text{utt})$$

$$\text{Literal listener} = \text{Binding convention} \times P(\text{ref})$$

To estimate the **colored parameters:**

Experiment 1 (N_{subj}=135, N_{item}=30): an antecedent selection task shows preferences and extent of preferences differ across items.



Experiment 2 (N_{subj}=28, N_{item}=30): a fill-in-the-blank task to test world knowledge bias.

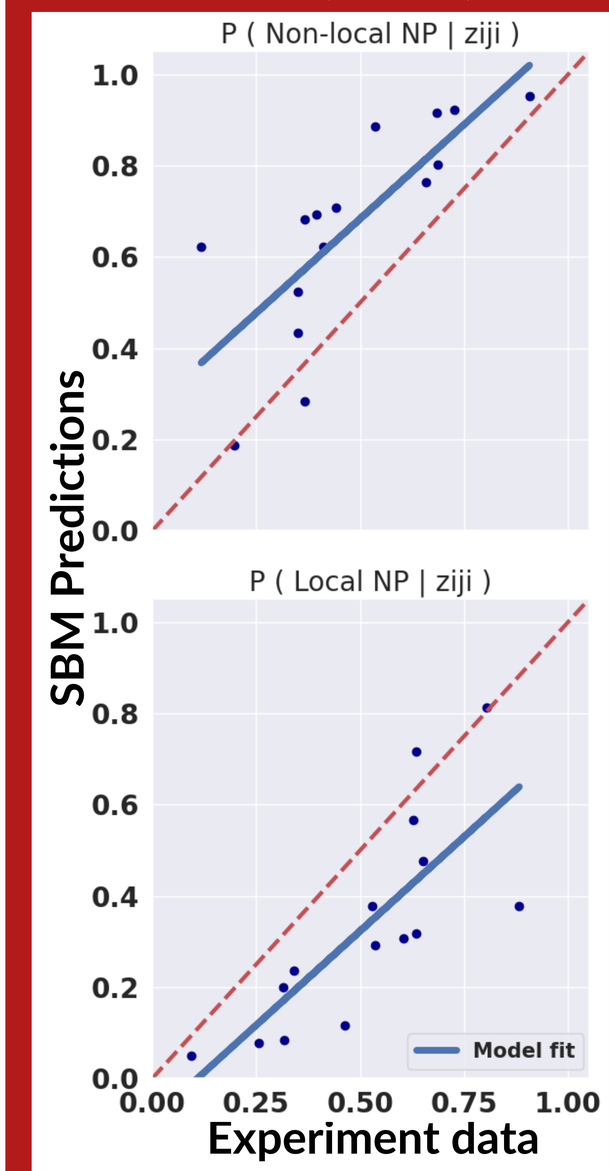
Experiment 3 (N_{subj}=65, N_{item}=30): a selection task, select a pronoun from a drop-down menu to convey a given referent.

Corpus Study [1]: **(Corpus of the Chinese Web)**

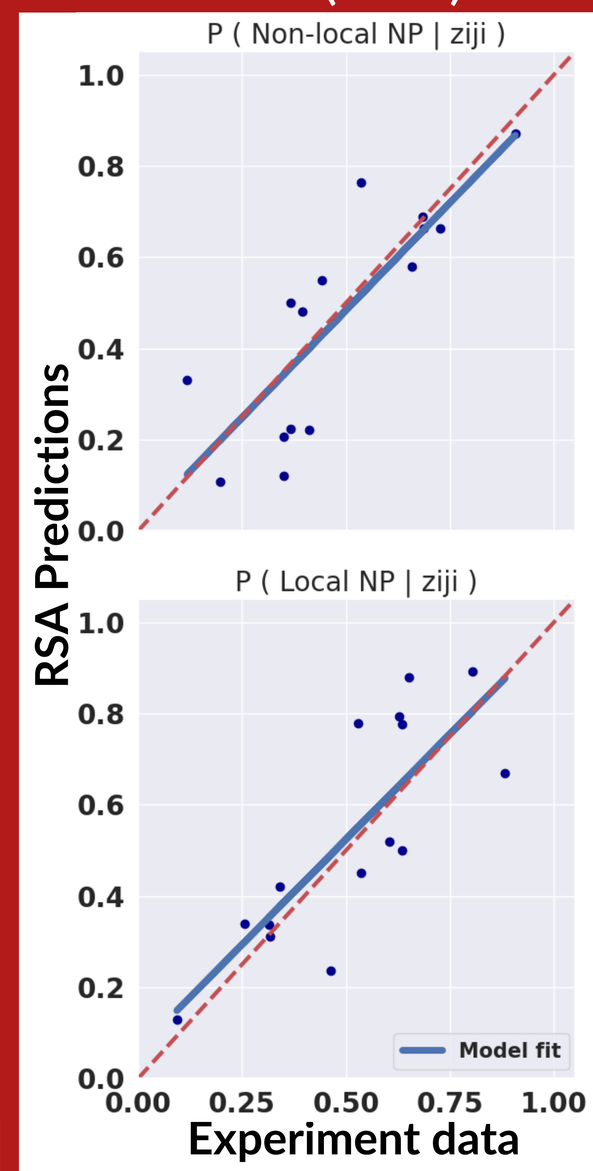
	ziji (self)	*ta* (him)	*taziji* (himself)
Freq	29.54%	**70.31%**	0.15%

Resolution of long-distance reflexive can be predicted by **Bayesian models.**

Simple Bayesian Model (SBM)



Rational Speech Act (RSA)



MODEL EVALUATIONS

	R^2	MSE	MAE	P-value
SBM	0.598	0.025	0.138	0.0007
RSA	**0.674**	**0.006**	**0.048**	0.0002

RESULTS

- **Item-by-item** quantitative analysis shows that **both SBM and RSA can make accurate predictions** for the resolution of ambiguous pronoun ziji.
- SBM **overestimates** the non-local antecedent choices and **underestimates** the local antecedent choices.

DISCUSSIONS

A **multilevel recursive reasoning** between listeners and speakers coupled with the **explicitly encoded frequency** of pronouns could improve Bayesian model predictions.

CONCLUSION

This study first shows interpretation biases for the within-sentence reflexive differ across items. We then provide a support case for Bayesian models of these non-uniform biases for within-sentence reflexive.

REFERENCES

[1] Schulz, M., Burnett, H., & Hemforth, B. (2021). A Journal of General Linguistics. [2] Zhan, M., Levy, R., Kehler, A. (2020). PLOS ONE. [3] Kehler, A, & Rohde, H. (2013). Theoretical Linguistics. [4] Frank, M. C., & Goodman, N. D. (2012). Science.

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